

Forward Flux Sampling for Tropical Cyclogenesis in a Neural Weather Emulator

John S. Schreck^{1,*}

¹National Center for Atmospheric Research, Boulder, CO

*Corresponding author: schreck@ucar.edu

April 6, 2026

Abstract

Tropical cyclone genesis is a rare atmospheric event whose probability varies by orders of magnitude across synoptic regimes, yet direct ensemble sampling cannot resolve this variability at operationally feasible ensemble sizes. This paper presents a *methods demonstration*: we show that Forward Flux Sampling (FFS), a non-equilibrium rare-event technique from statistical mechanics, can be efficiently coupled to a fast neural weather emulator (SDL-WXFormer, 1° resolution) to estimate conditional genesis rates without modifying model dynamics. The 1° emulator is chosen deliberately for computational tractability — FFS requires $\mathcal{O}(10^4)$ model trajectories per initial condition — and its calibrated stochastic layers provide the ensemble spread the method requires. FFS decomposes the rare $A \rightarrow B$ transition into a flux through an initial interface and a product of conditional crossing probabilities across four shooting interfaces, exploiting the natural parallelism of the shooting phase. Applied to 98 Atlantic basin initial conditions spanning August 21 – October 8, 2022, FFS resolves genesis rates spanning nearly three orders of magnitude (2.6×10^{-5} to 1.7×10^{-2} events day^{-1}), capturing a clear seasonal cycle whose timing and amplitude are qualitatively consistent with the known structure of the 2022 Atlantic season [National Hurricane Center, 2023] (suppressed early-season activity attributed to Saharan dust and wind shear, followed by a highly active September). A self-consistency check — comparing FFS rates to independent direct-sampling rates from the same flux trajectories — yields a mean ratio of 1.03 ± 0.15 (range 0.71–1.52) across all 98 initial conditions, confirming self-consistent implementation of the conditional probability factorization within this model. Computational enhancement factors range from $3\times$ (most active environment) to $140\times$ (most suppressed), with a geometric mean of $14\times$. Three case studies illustrate the physical diagnostics the method provides: the rate-limiting step is initial tropical organization for the Earl environment, uniformly high crossing probabilities for the Fiona precursor environment, and a compound barrier at the final intensification stages for the Ian environment. The methodology is independent of model architecture and extends directly to higher-resolution physics-based systems.

1 Introduction

Tropical cyclone genesis represents one of the most consequential rare events in atmospheric science. Approximately 10–15% of Atlantic tropical disturbances develop into named systems [Frank, 1970, Gray, 1968, Landsea, 1998], but discriminating developing from non-developing cases challenges deterministic and ensemble approaches alike [Titley and Elsberry, 2000, Halperin et al., 2013]. The challenge is not merely predictive: it is statistical. For a genesis event with probability $p \sim 10^{-3}$ per day, direct Monte Carlo estimation of the genesis rate to 20% relative uncertainty requires $\mathcal{O}(10^4)$ ensemble trajectories per initial condition — far beyond operational or research budgets when using physics-based models. The question of *how frequently* genesis occurs under a given atmospheric state, not just whether individual ensemble members develop, remains largely inaccessible to conventional approaches.

Rare event sampling techniques address this by intelligently concentrating computational effort along transition pathways. The broader rare event literature includes transition path sampling [Bolhuis et al., 2002] and metadynamics [Laio and Parrinello, 2002], which have primarily been developed for molecular and condensed-matter systems. The atmospheric community has primarily applied Adaptive Multilevel Splitting [AMS; Ragone et al., 2018, Webber et al., 2019, Gálfi et al., 2019] and related trajectory resampling methods [Finkel et al., 2023, 2024], which efficiently generate ensembles of rare *event realizations* and have been extended to heat waves, cold spells, and circulation extremes, including direct application to tropical cyclone intensification [Plotkin et al., 2019]. A parallel and rapidly developing line of work couples these ideas with machine learning emulators: Lancelin et al. [2025] recently demonstrated AI-boosted rare event sampling for characterizing extreme weather, establishing that learned surrogates can dramatically reduce the cost of importance sampling in atmospheric systems. Forward Flux Sampling [Allen et al., 2006, 2009, Valeriani et al., 2007, Hussain and Haji-Akbari, 2020] occupies a distinct but complementary niche: rather than generating rare realizations, FFS is designed specifically to estimate non-equilibrium steady-state *transition rates* — expected genesis events per unit time — with a factored uncertainty structure that identifies *which intensification stage is rate-limiting*. For operational genesis prediction, this rate-centric view is the natural target quantity.

FFS requires no equilibrium assumptions: no free energy landscapes, detailed balance, or Boltzmann distributions. The method needs only well-defined metastable regions, measurable flux through interfaces, and Markovian dynamics [Allen et al., 2009]. It has been applied across a broad range of non-equilibrium systems from biochemical networks [Allen et al., 2005] to nucleic acid folding kinetics [Schreck et al., 2015]. The atmosphere is a canonical non-equilibrium steady-state (NESS) system — driven persistently by solar forcing and dissipated by radiation and friction — and satisfies all three requirements. Atmospheric rare events are dynamical transitions between regions of a strange attractor [Lorenz, 1969, Touchette, 2009], not thermodynamic transitions over energy barriers, making FFS *more* defensible here than equilibrium-based rare event techniques.

An important interpretive point: FFS genesis rates vary by three orders of magnitude across the 98 initial conditions studied here, reflecting strongly non-stationary background conditions across

the hurricane season. The quantity k computed for each initial condition is a *conditional genesis rate given the atmospheric state at that IC*, not a climatological steady-state rate. Each FFS simulation treats a 10–15 day forecast window anchored to a specific ERA5 state as locally stationary, so k characterizes genesis probability density in that particular atmospheric environment. This conditional framing — analogous to a conditional return period given present climate state [Finkel et al., 2023, 2024] — motivates our seasonal sampling across 98 ICs and enables direct interpretation against observed Atlantic activity.

Machine learning weather models [Lam and Coauthors, 2023, Bi and Coauthors, 2023, Pathak and Coauthors, 2022] offer a natural platform for rare event sampling: they are $\sim 1000\times$ faster than physics-based models, natively generate ensemble spread through stochastic parameterizations, and inherit the dynamical attractor of their training reanalysis. This speed advantage is decisive for FFS, which requires $\mathcal{O}(10^4)$ trajectories per initial condition. Recent work has begun exploiting this combination for subseasonal-to-seasonal extreme event forecasting [Mu et al., 2025] and importance sampling of weather extremes [Lancelin et al., 2025]. The SDL-WXFormer [Schreck and Coauthors, 2025] employed here extends the standard deterministic emulator with stochastic decomposition layers calibrated to ERA5 ensemble statistics, providing the trajectory diversity the FFS shooting phase requires.

Scope and model choice. This paper is primarily a *methods demonstration*: we establish that FFS can be coupled to a fast neural emulator to produce physically interpretable genesis rate estimates, and characterize the method’s behaviour across a full hurricane season. We deliberately use SDL-WXFormer at 1° resolution rather than a convection-permitting physics-based model for two practical reasons. First, FFS is computationally demanding: each initial condition requires $\mathcal{O}(10^4)$ 15-day trajectories, and at ~ 1.5 GPU-minutes per forecast on an NVIDIA A100, the full 98-IC dataset required approximately 2,450 GPU-hours. A comparable experiment with WRF or IFS would be prohibitively expensive on available allocations. Second, the SDL-WXFormer’s calibrated stochastic decomposition layers provide the ensemble variability the FFS shooting phase requires; a deterministic neural emulator would collapse all branching probabilities to zero or one. The 1° resolution imposes a well-understood limitation: the model cannot faithfully reproduce Category 3+ intensity or convective-scale vortex dynamics, so state B is defined at 975 hPa (tropical storm to minimal hurricane) rather than the intensities typical of major hurricane development. This limitation is consistent with the paper’s methodological focus, and the framework transfers directly to higher-resolution systems as compute budgets permit.

This paper reports: (1) a three-order-of-magnitude seasonal cycle in conditional genesis rates qualitatively consistent with the known structure of the 2022 Atlantic season; (2) a self-consistency validation showing FFS and direct-sampling rates agree within a mean ratio of 1.03 ± 0.15 across 98 initial conditions; (3) case studies illustrating physically distinct rate-limiting steps; and (4) a Cyclone Phase Space contamination screen separating tropical from extratropical development pathways.

2 Methods

2.1 Forward Flux Sampling

FFS estimates the transition rate from metastable state A (unorganized disturbances, $Q > \lambda_A$) to state B (organized tropical cyclone, $Q \leq \lambda_B$) by factorizing:

$$k_{A \rightarrow B}^{\text{FFS}} = \Phi_{A,0} \times \prod_{i=1}^n P(\lambda_i | \lambda_{i-1}) \quad (1)$$

where $\Phi_{A,0} = N_{\text{cross}}/T_{\text{total}}$ is the flux of trajectories crossing the first interface λ_0 per unit time, and $P(\lambda_i | \lambda_{i-1})$ is the conditional probability that a trajectory at λ_{i-1} reaches λ_i before returning to A (Fig. 1). The order parameter $Q(\mathbf{x})$ is the minimum sea-level pressure (MSLP) within the Atlantic basin domain $\Omega = [10^\circ\text{N}, 40^\circ\text{N}] \times [100^\circ\text{W}, 20^\circ\text{W}]$.

Order parameter justification. MSLP is the community-standard metric for tropical cyclone intensity and is used operationally by NHC, JTWC, and all major forecast centers to define genesis and track intensity throughout a storm’s lifecycle [Knaff and Zehr, 2007]. It has well-understood properties at 1° resolution (see state- B threshold discussion below), and is robustly trackable without requiring multi-variable composites that would introduce additional calibration uncertainty. We acknowledge that MSLP is a downstream integrator of the thermodynamic and kinematic conditions that enable genesis, rather than a direct measure of those precursors. However, for the purpose of defining the FFS interfaces and measuring crossing probabilities, MSLP’s role as a monotone descriptor of vortex organization is well suited: the seasonal cycle resolved by the FFS rates (Figure 2) is physically coherent, and the geographic distribution of state- B arrivals (Figure 6) clusters in the observed Atlantic main development region. Future implementations could extend the order parameter to incorporate low-level vorticity, mid-tropospheric moisture, or a composite genesis potential index [e.g., Bister and Emanuel, 1998, Camargo et al., 2007]; such extensions are expected to sharpen the physical interpretability of per-interface bottleneck diagnostics but are not required to demonstrate that FFS produces internally consistent rate estimates, which is the goal of this paper.

State definitions and interfaces. State A : $Q > \lambda_A = 1008$ hPa (unorganized synoptic disturbances). State B : $Q \leq \lambda_B = 975$ hPa AND warm-core structure satisfied (CPS criterion) AND vortex centre within the basin domain Ω . We note that 975 hPa corresponds to tropical storm to minimal hurricane intensity and is *intentionally stronger* than the conventional NHC tropical storm genesis designation (~ 34 kt, roughly 997–1003 hPa in typical Atlantic pressure–wind relationships). At 1° resolution, the model’s smoothed MSLP field does not develop a reliable closed isobar signature until the vortex is more strongly organised than the standard genesis threshold; 975 hPa is the lowest pressure at which state- B arrivals cluster in physically expected geographic locations and pass the CPS warm-core screen with consistent reliability. The compound criterion means that reaching the 975 hPa MSLP threshold is necessary but not sufficient for state B ; trajectories that deepen to 975 hPa while recurving poleward or exiting the domain are not counted as

state- B arrivals, and $P_4 = P(B|\lambda_4) < 1$ reflects this screening. The 975 hPa threshold is calibrated to the SDL-WXFormer’s 1° horizontal resolution through trial experiments (see Appendix A). The flux interface and four shooting interfaces are:

$$\lambda_0 = 1000.0, \quad \lambda_1 = 987.2, \quad \lambda_2 = 984.7, \quad \lambda_3 = 981.2, \quad \lambda_4 = \lambda_B = 975.0 \quad [\text{hPa}]$$

satisfying $\lambda_A > \lambda_0 > \lambda_1 > \dots > \lambda_4 = \lambda_B$. The interfaces and thresholds stated here are those used to produce all results in this paper and match the configuration file in the public code repository (`config/ffs.yml`; <https://github.com/NCAR/miles-tails>). Note that `forecast_start_times` in that file lists only the three case-study ICs; the full 98-IC seasonal run was launched via the PBS job scripts in `applications/run_paper_figures_mar18.sh`.

Notation. We label the four conditional crossing probabilities P_1 – P_4 in order of the FFS product: $P_1 = P(\lambda_2|\lambda_1)$, $P_2 = P(\lambda_3|\lambda_2)$, $P_3 = P(\lambda_4|\lambda_3)$, and $P_4 = P(B|\lambda_4)$. The $\lambda_0 \rightarrow \lambda_1$ transition is implicit in the flux $\Phi_{A,0}$ and is not a shooting interface; P_1 therefore refers to the *first shooting step* ($987.2 \rightarrow 984.7$ hPa), not the first interface crossing from A .

Interface placement. Shooting interfaces are chosen using the optimal- $1/e$ rule of Allen et al. [2009]: the next interface λ_{i+1} is placed at the p^* -quantile of the MSLP distribution reached by trajectories launched from λ_i , where $p^* = 1/e \approx 0.368$. This placement minimises variance in the estimated rate and targets conditional crossing probabilities near $1/e$, maximising statistical efficiency [Allen et al., 2009]. In practice, interfaces were optimized from long trial runs and then rounded to physically interpretable pressure values; the resulting P_i values (Table 1) cluster between 0.10 and 0.91 across all ICs, with a majority in the well-conditioned range 0.25–0.75 as shown in Figure 5.

Flux estimation. For each initial condition we generate trajectories from the ERA5 state perturbed by SDL-WXFormer stochastic layers, advancing 6-hourly for up to 15 days. Decorrelated λ_0 crossings (requiring return to state A between consecutive crossings) are accumulated to $N_{\text{cross}} = 2000$ per initial condition, yielding Poisson uncertainty $\sigma_\Phi/\Phi \approx 2.2\%$ — well below across-IC variability. State- B arrivals during flux generation provide an independent direct-sampling rate $k^{\text{direct}} = N_{\text{direct}}/T_{\text{total}}$ that serves as a self-consistency check.

Shooting. Configurations saved at λ_0 seed shooting trajectories toward each successive interface. For each interface λ_{i-1} , trajectories are launched until 2000 successful crossings of λ_i are accumulated. Success probability $P(\lambda_i|\lambda_{i-1}) = N_{\text{success}}/N_{\text{attempts}}$ with binomial uncertainty. Statistical uncertainty on the full rate (Eq. 1) propagates from each factor:

$$\frac{\sigma_k}{k} = \sqrt{\left(\frac{\sigma_\Phi}{\Phi}\right)^2 + \sum_{i=1}^n \left(\frac{\sigma_{P_i}}{P_i}\right)^2} \quad (2)$$

2.2 SDL-WXFormer and Experimental Setup

SDL-WXFormer [Schreck and Coauthors, 2025] is a vision transformer trained on ERA5 reanalysis [Hersbach et al., 2020] (1979–2021) at 1° resolution, 6-hour timesteps, predicting 69 pressure-level and 8 surface variables. Stochastic decomposition layers inject learned, state-dependent noise enabling calibrated ensemble generation. Each 10-day forecast requires ≈ 1.5 GPU-minutes on NVIDIA A100.

We select 98 initial conditions from ERA5 at 00 and 12 UTC spanning August 21 – October 8, 2022, verifying $Q(\mathbf{x}_0) > 1008$ hPa for each. Three initial conditions receive detailed case study treatment:

- **2022-09-02T00Z (Earl)**: ~ 24 hours before Hurricane Earl’s official tropical-storm genesis on September 3.
- **2022-09-09T12Z (Fiona)**: highest observed FFS genesis rate in the dataset, within the Fiona precursor environment (~ 5 days before Fiona’s formation).
- **2022-09-22T00Z (Ian)**: 24 hours before Hurricane Ian’s NHC genesis designation on September 23.

2.3 Feature Tracking and Tropical Contamination Screen

We emphasise at the outset that both the vortex tracker and the warm-core screen described below are purpose-built, ad-hoc components calibrated for the SDL-WXFormer at 1° resolution. They were sufficient to demonstrate that FFS produces self-consistent genesis rates in this setting, which is the goal of this methods demonstration. Production-quality implementations at other resolutions or with other models will require state-of-the-art objective trackers (e.g., TempestExtremes, TRACK, tobac [Heikenfeld and Coauthors, 2019], or similar) and a formally validated tropical classification scheme; the present implementations should not be adopted uncritically for other applications.

Flux mode (multi-storm tracking). During the flux phase, the order parameter Q is the global minimum MSLP over the basin domain Ω , which may correspond to any co-existing tropical system. To track this minimum coherently across timesteps, we maintain a running list of all active vortex candidates. At each 6-hour step, Gaussian smoothing ($\sigma = 1.5$ grid-points) is applied to the MSLP field and local minima below 1010 hPa are detected. Existing tracks are updated by nearest-neighbour matching within 12° ; unmatched minima below 12°N over land are suppressed (terrain artefacts); remaining unmatched minima seed new tracks. A track is retained for up to two consecutive timesteps of failed matching before being dropped. This scheme simultaneously follows multiple co-existing systems and switches the designated “minimum” when a new system deepens below the current one, preserving a well-defined monotone order parameter throughout the flux phase.

Shoot mode (single-vortex tracking). Once a trajectory has been saved at λ_0 and begins the shooting phase, the target vortex is the one that triggered the λ_0 crossing. At each timestep,

the MSLP minimum is sought within 12° of the vortex’s last known position. This single-vortex constraint prevents Q from jumping to a stronger but unrelated system, ensuring that the shooting trajectory measures the intensification of the specific disturbance seeded at λ_0 .

Tropical contamination screen. An empirically-calibrated contamination screen, inspired by the Cyclone Phase Space framework of Hart [2003], suppresses two known sources of spurious state- B arrivals at 1° resolution: (1) terrain-induced MSLP minima from hydrostatic extrapolation errors, and (2) mid-latitude baroclinic cyclones entering the domain. The screen evaluates the thermal wind parameters $-V_T^L$ and $-V_T^U$ at λ_0 crossings and when a tracked vortex exceeds 50°N or -10°W , rejecting systems with $-V_T^L > 0$ or $-V_T^U > 0$. The rejection thresholds were calibrated by manual inspection of a subset of trajectories in the 2022 dataset that were visually identifiable as extratropical (recurving systems and baroclinic cyclones intruding from the northwest); they are not derived from formal CPS validation at this resolution and should be treated as engineering choices rather than physical thresholds. The screen errs toward inclusion to avoid discarding ambiguous cases.

Across the full dataset, rejection rates at each interface are approximately 30% (λ_1), 20% (λ_2), 27% (λ_3), and 43% (λ_4), with per-IC variation from 23% to 93% at λ_4 in the most suppressed environments. The elevated rejection rate at λ_4 is physically expected: 975 hPa is precisely the pressure range where midlatitude baroclinic systems deepen most strongly, so contamination from extratropical tracks is worst at the final interface. The non-monotone pattern at earlier interfaces reflects marginal cyclones that failed to maintain a warm core through the intensification cascade. The high per-IC variation at λ_4 (up to 93% in suppressed environments) reflects that those rare vortices which deepen to 975 hPa in unfavorable environments are disproportionately anomalous poleward-recurving tracks rather than canonical tropical development.

3 Results

3.1 Seasonal Genesis Rates and Self-Consistency Check

Figure 2 shows FFS genesis rates across all 98 initial conditions. The full seasonal cycle spans nearly three orders of magnitude, from $2.6 \times 10^{-5} \text{ day}^{-1}$ during suppressed October conditions to $1.7 \times 10^{-2} \text{ day}^{-1}$ during the peak-activity period around September 9–10 (consistent with the Fiona formation environment). This dynamic range is entirely unresolvable by direct 50-member ensemble sampling. For context, the ECMWF IFS 50-member ensemble [Buizza and Coauthors, 2018] analyzed over the same 98 initial conditions reaches the minimum MSLP threshold in 25.4% of members on average; this figure is included in Figure 2 as a *qualitative sanity check* only. A direct rate comparison between FFS (SDL-WXFormer, 1°) and IFS (0.25° physics, 1.5° output) is not appropriate: IFS does not apply the decorrelated-crossing requirement defining λ_0 flux, does not use a CPS filter, and operates at different resolution with different physics. The IFS figures confirm that strong MSLP deepening occurs in the model at a plausible frequency; quantitative rate comparisons between the two systems are deferred to future work with matched configurations.

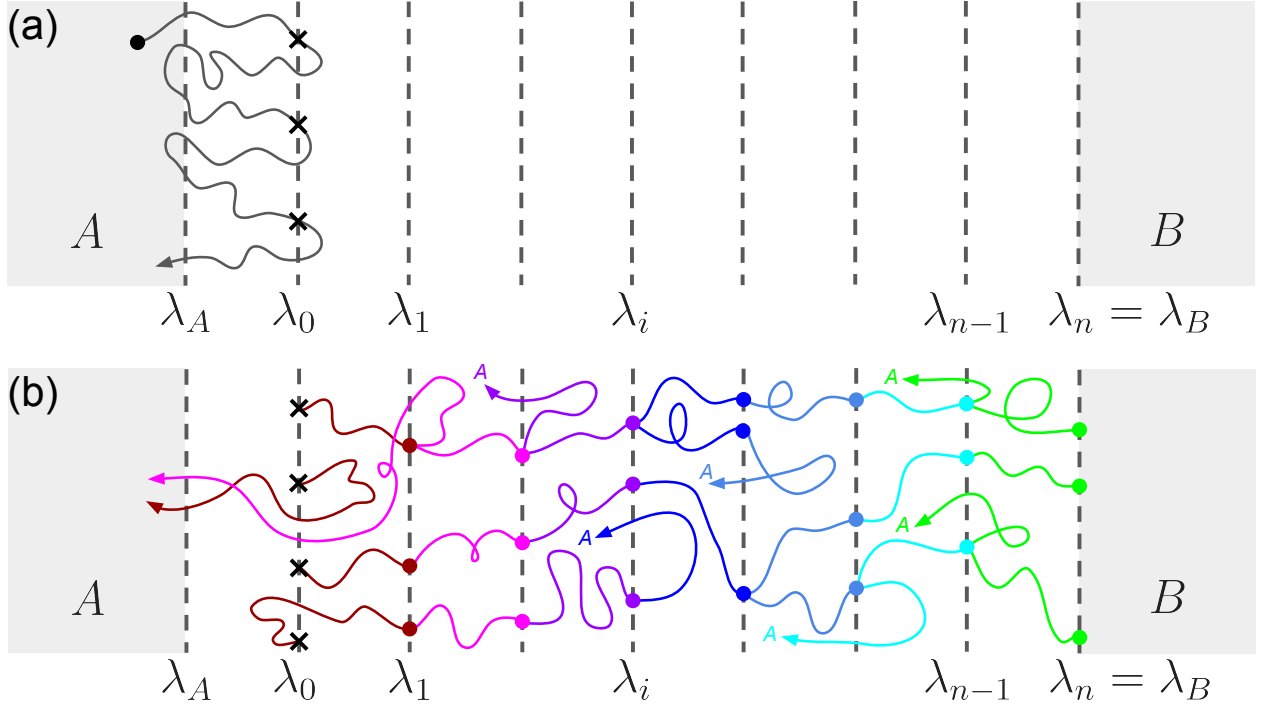


Figure 1: Schematic of the Forward Flux Sampling algorithm (after Allen et al. 2006). *Panel (a)* — *flux phase*: Gray trajectories evolve forward from state A ; each decorrelated crossing of λ_0 is recorded (marked $\times$). *Panel (b)* — *shooting phase*: New trial trajectories are launched from each saved λ_i configuration (filled circles). Shots that return to state A (arrows labelled A) are failures; shots that reach λ_{i+1} (filled circles on the next interface) are successes, yielding the conditional probability $P(\lambda_{i+1}|\lambda_i)$. The genesis rate is $k = \Phi_{A,0} \prod_i P(\lambda_{i+1}|\lambda_i)$.

The seasonal evolution of each individual conditional probability P_i (Fig. 3) shows that P_1 varies most strongly with the seasonal cycle, while P_4 is the most uniformly suppressed across all ICs.

The primary self-consistency check is the agreement between k^{FFS} and the independent direct-sampling rate k^{direct} (Figure 4). These estimates are derived from entirely separate components of the same simulation: k^{FFS} is a product of flux and shooting probabilities, while k^{direct} counts raw state- B arrivals during the flux phase. Across all 98 initial conditions, the ratio $k^{\text{FFS}}/k^{\text{direct}} = 1.03 \pm 0.15$ (mean $\pm$ std; median 1.02; range 0.71–1.52; Figure 4). The near-unity mean and median indicate no systematic bias in the factorization. The standard deviation of 0.15 and the factor-of-two range reflect the Poisson sampling noise inherent in k^{direct} (which has few counts at suppressed ICs), not a bias in k^{FFS} . Critically, this agreement holds across three orders of magnitude of genesis probability, which would be expected only if (1) the interface spacing produces well-conditioned probability estimates, (2) the ensemble sizes are adequate, and (3) the CPS screen does not introduce systematic bias. We interpret this agreement as confirming the correct implementation of the conditional probability factorization within the SDL-WXFormer framework. It is important to note what this check does and does not establish. It confirms that the FFS bookkeeping arithmetic is internally consistent: the product $\Phi_{A,0} \prod P_i$ recovers the same rate that direct state- B counting yields. It does *not* constitute verification against observed climatology, nor does it validate the Markov assumption (see below), nor does it guarantee that the absolute rate values are unbiased with respect to the real atmosphere.

Markovianity. FFS requires that a trajectory’s future behaviour depends only on its current state, not on its history prior to the most recent interface crossing. For SDL-WXFormer, this question has two distinct dimensions.

Architectural Markovianity. Enabling rare-event sampling via FFS was an explicit design objective of SDL-WXFormer, and the model was architected as a strict state-in/state-out emulator specifically to satisfy the Markov requirement. The base deterministic WXFormer is made stochastic by adding Stochastic Decomposition Layer (SDL) modules and fine-tuning with a CRPS loss [Schreck and Coauthors, 2025]. At inference, each 6-hour forward step takes only the current atmospheric state vector as input and applies the learned weights plus an independent stochastic perturbation drawn from the SDL layers; no recurrent memory is carried between steps. This design means the model is Markovian at inference by construction, to the extent permitted by the available training data and the choice of state variables included.

Physical Markovianity and state-vector completeness. Whether the FFS rate calculation is physically Markovian is a separate question that hinges on whether the model’s state vector is sufficient to capture all variables that influence genesis probability. Variables absent from the state vector represent the primary source of implicit non-Markovian pressure. SDL-WXFormer does not explicitly predict slow boundary-condition fields such as sea-surface temperature (SST) or soil moisture; these are held fixed at their initialisation values rather than evolving during the simulation. However, the model’s weights encode multi-step climatological relationships between these slow variables and atmospheric dynamics learned during training. Trajectories launched from

λ_i over anomalously warm SST, or from environments with anomalous soil moisture, will therefore evolve differently in ways not captured by the atmospheric state vector alone. To the extent that these boundary-condition fields at initialisation are correlated with the subsequent evolution encoded in the weights, this introduces a background-state dependence that a fully Markovian FFS would not carry. We note that the observed anti-correlation between P_1 and P_4 across ICs (environments favorable for initial organisation tend to have suppressed final-stage probabilities) reflects *cross-IC* atmospheric and SST variability, not within-trajectory memory, and does not itself violate the Markov assumption. The self-consistency check can detect Markov violations that produce systematic bias in the factorized rate relative to the direct count, but cannot detect violations that bias both estimators in the same direction. A more rigorous test — verifying that $P(B|\lambda_i)$ depends only on the current state and not on SST or other omitted variables — is deferred to future work.

Computational enhancement factors range from $3\times$ (most active IC) to $140\times$ (most suppressed), with geometric mean $14\times$.

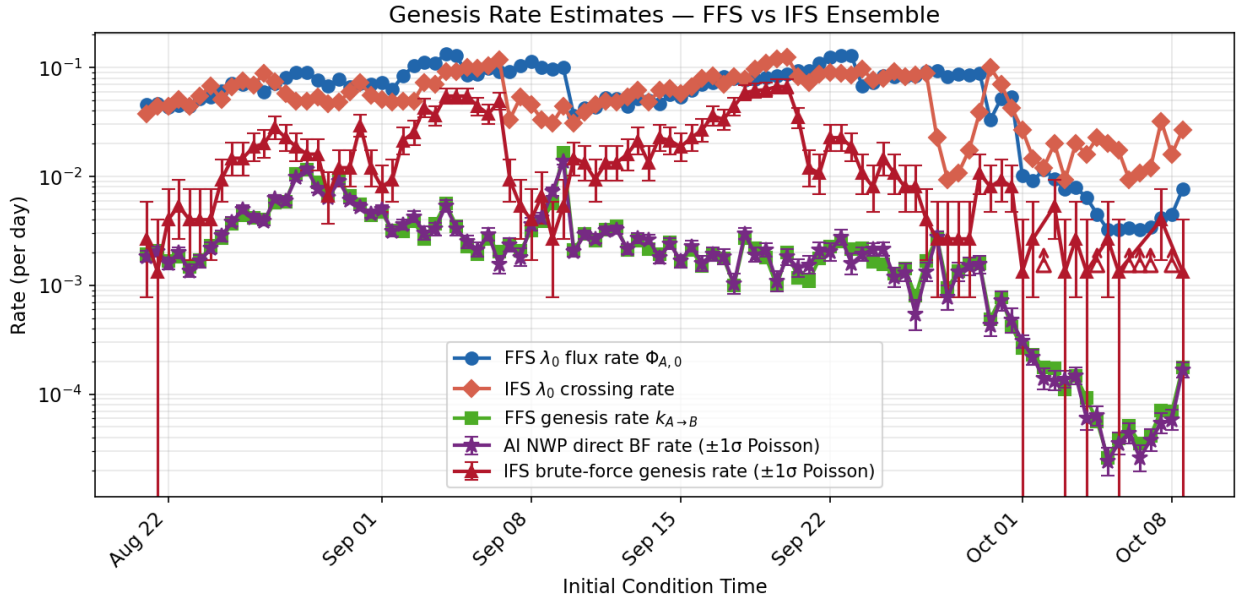


Figure 2: FFS genesis rates across all 98 initial conditions (August 21 – October 8, 2022) on a logarithmic scale. *Blue*: λ_0 flux rate $\Phi_{A,0}$. *Orange*: IFS λ_0 crossing rate (contextual reference; not directly comparable to FFS due to different resolution and no CPS filter). *Green*: Full FFS genesis rate k^{FFS} (flux $\times$ product of conditional probabilities). *Purple*: Direct-sampling rate k^{direct} from state- B arrivals during the flux phase ($\pm 1\sigma$ Poisson). *Red*: IFS brute-force genesis rate ($\pm 1\sigma$ Poisson; upward arrows denote upper bounds where zero events observed). Agreement between green and purple (mean ratio 1.03 ± 0.15) validates the FFS method. The FFS rate resolves a seasonal cycle spanning nearly three orders of magnitude invisible to direct 50-member ensemble sampling.

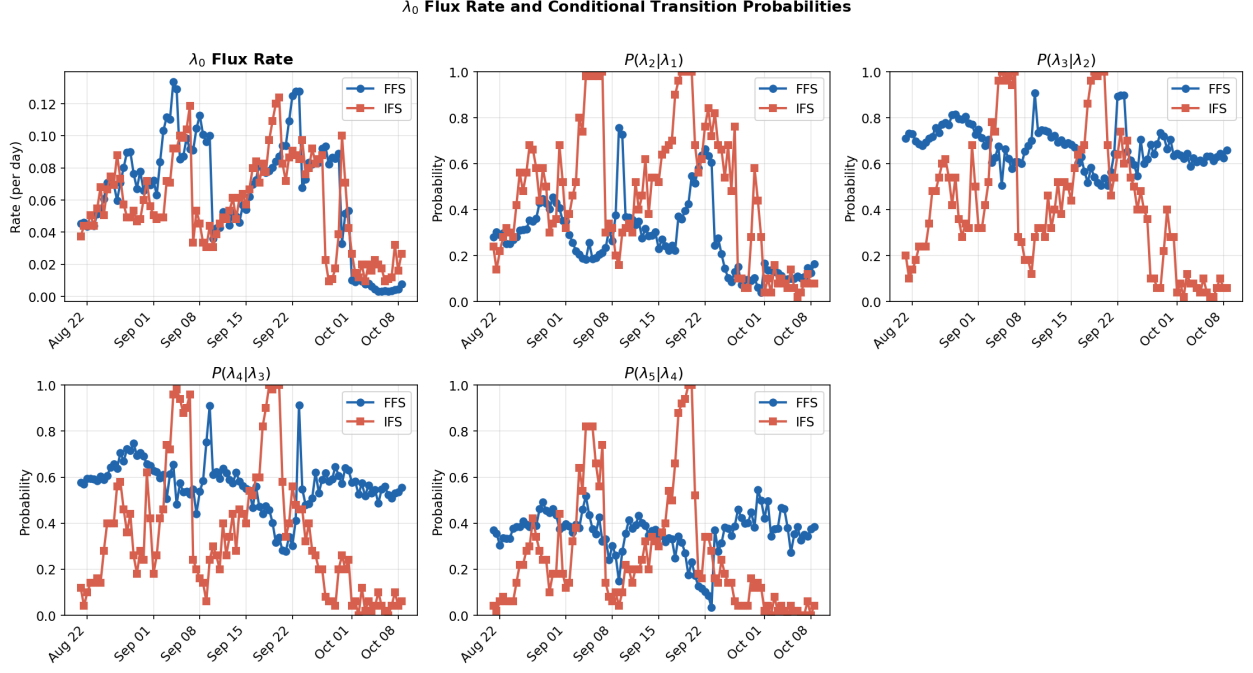


Figure 3: Seasonal evolution of FFS conditional crossing probabilities $P(\lambda_i|\lambda_{i-1})$ for all 98 initial conditions (August 21 – October 8, 2022). The first panel shows the λ_0 flux rate; subsequent panels show each P_i between successive shooting interfaces. The dashed line marks the all-season mean. P_1 (initial organization, $\lambda_1 \rightarrow \lambda_2$) varies most strongly with the seasonal cycle, with a pronounced peak in early September coinciding with the Fiona formation environment. P_4 (final intensification, $\lambda_4 \rightarrow B$) is the most uniformly suppressed probability, limiting the overall genesis rate even in active periods. The three case-study ICs (Earl 09-02, Fiona 09-09, Ian 09-22) are indicated by vertical markers.

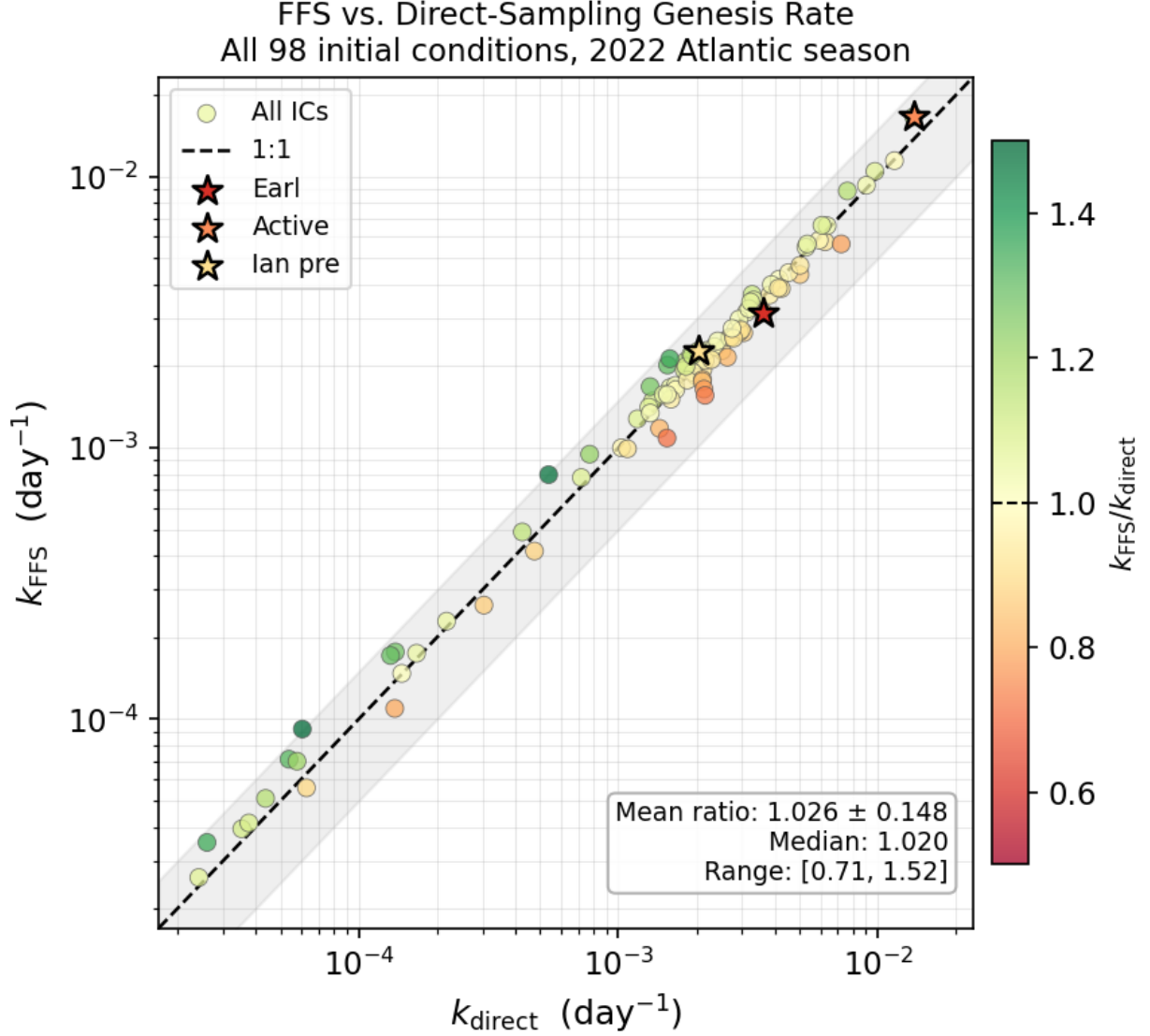


Figure 4: Log-log scatter of FFS genesis rate k^{FFS} versus independent direct-sampling rate k^{direct} for all 98 initial conditions. Points are coloured by the ratio $k^{\text{FFS}}/k^{\text{direct}}$: green indicates near-unity agreement; red/blue indicate over- or under-estimation. The black dashed line is the 1:1 reference; shaded band spans $\pm 50\%$. Three case-study ICs are marked with stars (Earl: red; Fiona: orange; Ian: yellow). The tight clustering along the 1:1 line across three orders of magnitude confirms correct implementation of the conditional probability factorization. Inset statistics: mean ratio 1.03 ± 0.15 , median 1.02, range [0.71, 1.52].

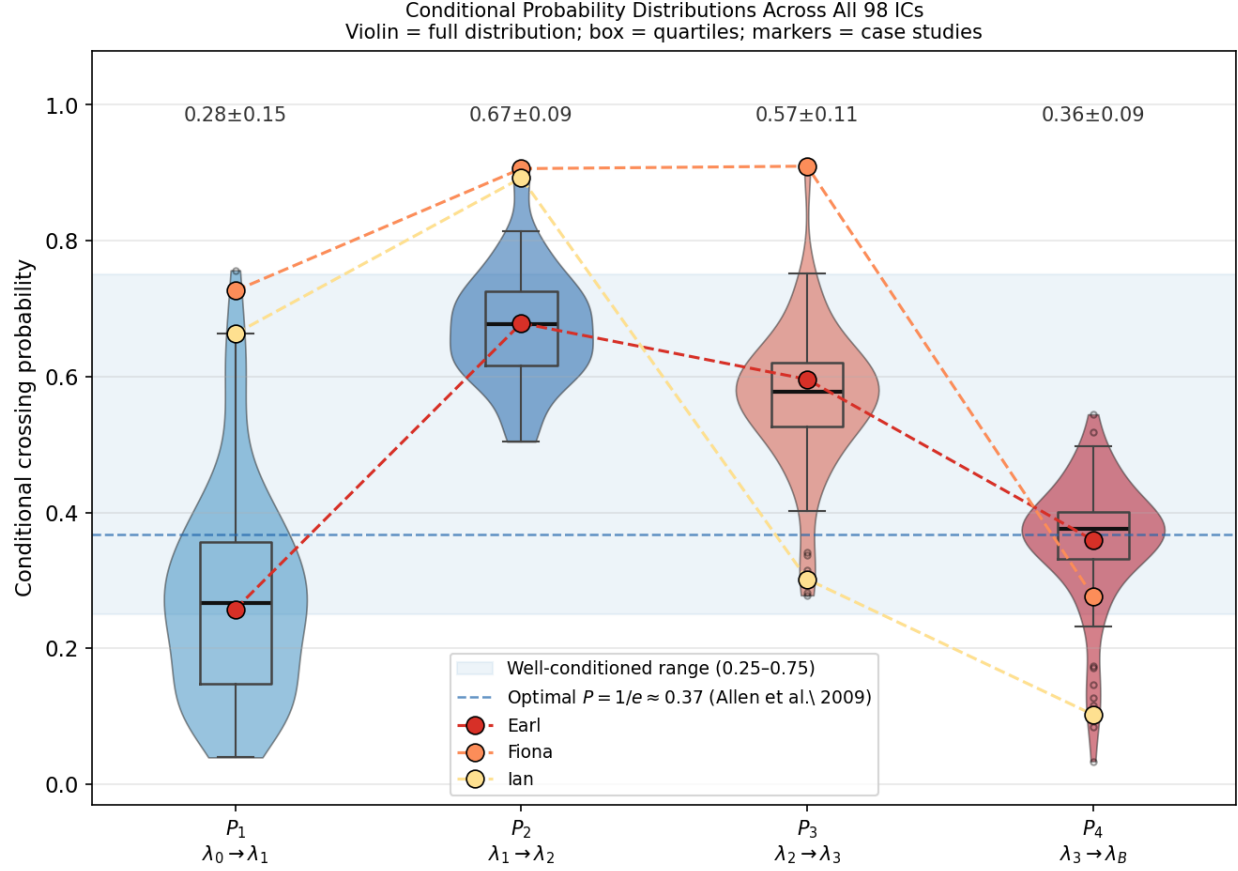


Figure 5: Distribution of conditional crossing probabilities P_i across all 98 initial conditions. Each violin shows the full distribution; boxes show quartiles; coloured lines connect the three case-study ICs. The horizontal dashed line marks the optimal $P = 1/e \approx 0.37$ [Allen et al., 2009]; the blue band spans the well-conditioned range 0.25–0.75. Most ICs have P_i within the well-conditioned range, confirming that interface placement produces efficient variance reduction. The spread of P_1 (initial organization) is larger than that of P_2 – P_3 , reflecting the dominant role of large-scale atmospheric variability in setting the first intensification barrier. Mean $\pm$ SD for each interface are annotated.

3.2 Case Studies: Distinct Genesis Bottlenecks

Table 1 presents FFS statistics for the three case-study initial conditions. The most striking feature is the variable position of the rate-limiting step, encoded in which conditional probability is smallest. Figure 6 shows the geographic distribution of state- B arrivals across all 98 ICs, confirming that the model’s genesis events cluster in physically plausible locations (tropical Atlantic, Caribbean, Gulf of Mexico) and shift geographically with the seasonal cycle. For the Earl IC specifically, Fig. 7 shows the 199 reactive pathways that successfully reached state B , illustrating the geographic diversity of intensification trajectories sampled by FFS from a single background state.

Table 1: FFS statistics for three case-study initial conditions. $\Phi_{A,0}$: flux rate. P_i : conditional crossing probability at interface λ_i . k^{FFS} : full genesis rate (Eq. 1). k^{direct} : independent direct-sampling rate. Ratio: $k^{\text{FFS}}/k^{\text{direct}}$. Speedup: enhancement over direct sampling for equivalent 20% rate uncertainty. Bottom row: statistics across all 98 ICs.

IC	$\Phi_{A,0}$ (day ⁻¹)	P_1	P_2	P_3	P_4	k^{FFS} (day ⁻¹)	k^{direct} (day ⁻¹)	Ratio	Speedup
Earl (09-02T00Z)	0.0837	0.258	0.679	0.596	0.359	3.13×10^{-3}	3.61×10^{-3}	0.87	10×
Fiona (09-09T12Z)	0.1002	0.726	0.906	0.910	0.276	1.66×10^{-2}	1.37×10^{-2}	1.21	3×
Ian (09-22T00Z)	0.1247	0.663	0.893	0.302	0.102	2.28×10^{-3}	2.04×10^{-3}	1.12	19×
All-98 mean±SD	0.065 ± 0.034	0.278 ± 0.154	0.674 ± 0.086	0.569 ± 0.107	0.356 ± 0.093	2.6×10^{-3}	2.5×10^{-3}	1.03 ± 0.15	14× [†]

[†]Geometric mean; range 3–140×. Speedup distribution is right-skewed; geometric mean is the appropriate summary.

For the **Earl IC**, $P_1 = 0.258$ is the dominant bottleneck: fewer than 1 in 4 trajectories crossing 1000 hPa organize to 987 hPa in this environment. This is physically consistent with the pre-genesis atmospheric state ~24 hours before Earl’s classification, when the system had established a coherent low-level circulation but had not yet achieved the sustained convective organization needed to cross the first intensification threshold.

For the **Fiona IC** (September 9), all four conditional probabilities are elevated (≥ 0.27), and the product $\prod P_i = 0.166$ is ~4–9× larger than the other two cases. The dominant bottleneck shifts to $P_4 = 0.276$ (final intensification to 975 hPa), reflecting an environment that readily organizes disturbances but where crossing the hurricane-threshold step remains non-trivial. The FFS rate (1.66×10^{-2} day⁻¹) is high enough that direct sampling is also feasible (speedup 3×), providing a strong cross-validation: both methods agree to within 21% for this most-active IC.

The **Ian IC** presents the most physically interesting pattern. $P_1 = 0.663$ and $P_2 = 0.893$ indicate that the environment readily supports initial organization through 985 hPa, consistent with the warm, moist, low-shear conditions present over the Caribbean and western Atlantic on September 22. But $P_3 = 0.302$ and $P_4 = 0.102$ reveal a hard barrier in the final intensification steps (981 hPa to 975 hPa). This is physically consistent with Ian’s trajectory: the storm was still organizing over the Caribbean on September 22–23, entering the Gulf of Mexico’s warm core environment only after September 25, when conditions supported rapid intensification. The FFS framework identifies the final-stage intensification barrier *one day before* Ian’s official genesis — a diagnostic that direct ensemble sampling cannot resolve.

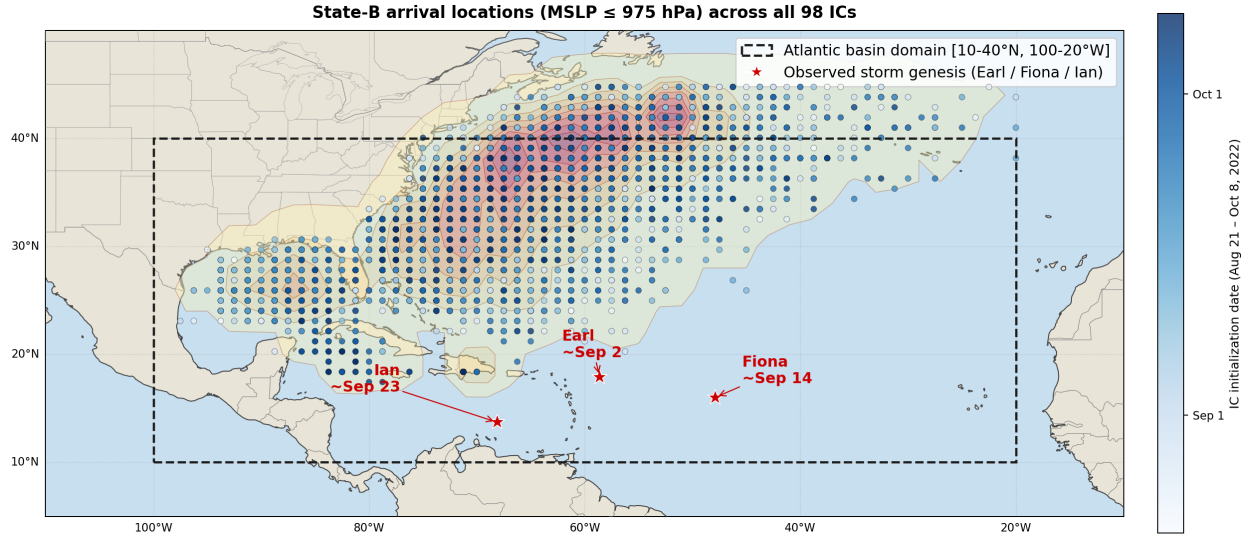


Figure 6: Geographic distribution of state- B arrival locations ($\text{MSLP} \leq 975 \text{ hPa}$) across all 98 initial conditions. Each point marks where a reactive trajectory first reached state B ; colour encodes the IC initialisation date from August 21 (light) to October 8 (dark). The 2° density contours (background) show the spatial clustering of genesis events across the season. Approximate observed genesis locations for Earl, Fiona, and Ian are annotated for reference. The distribution is concentrated in the tropical Atlantic, Caribbean, and Gulf of Mexico, consistent with observed 2022 Atlantic storm tracks, and shifts westward later in the season as Ian-like Gulf environments become active.

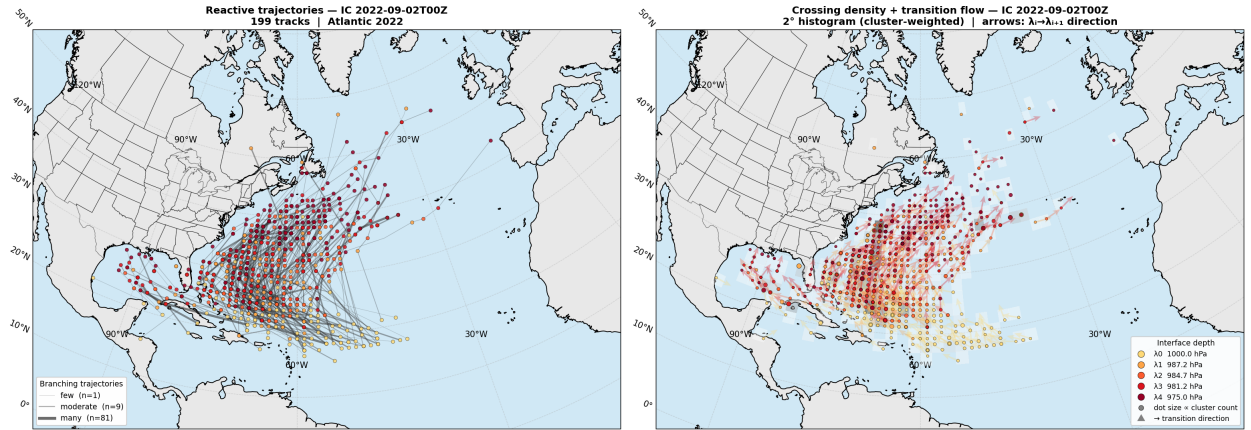


Figure 7: Reactive genesis trajectories for the Earl IC (2022-09-02T00Z). *Left*: Individual tracks that cross from λ_0 through to state B , color-coded by branching factor (cluster weight); the 199 reactive pathways sample a broad region spanning the tropical Atlantic, Caribbean, and western subtropical Atlantic. *Right*: Cluster-weighted crossing density (2° histogram) and mean $\lambda_n \rightarrow \lambda_{n+1}$ transition direction (arrows), revealing the geographic regions most active in the intensification cascade for this environment. CPS-rejected (extratropical) trajectories are excluded from both panels.

3.3 FFS Shooting Trees

Figure 8 shows representative FFS shooting trees for each of the three case-study ICs. Each tree traces the branching genealogy of a single λ_0 seed: the star marks the seed location, and successive interface crossings fan out as the ensemble of reactive pathways progresses toward state B . The geographic spread of λ_4 descendants reflects the diversity of intensification pathways sampled by the FFS from a single starting configuration.

Importantly, the IC date labels are *initialisation* times, not genesis dates. Each FFS simulation runs 15 days forward from its IC; each node in the tree represents the storm feature at a different future lead time within that window. A single seed at λ_0 therefore spawns descendants that span a range of calendar dates, geographic positions, and atmospheric states — all sampled from one background IC.

All three trees are displayed on a single combined map; branch colours encode interface level (λ_1 through λ_4) using a shared yellow-orange-red scale, and each IC is identified by a text label. The Earl tree (2022-09-02T00Z, 24 hours before Earl’s genesis) seeds near 18°N in the eastern Atlantic and fans westward, consistent with Earl’s observed west-northwest track toward the Leeward Islands. The Fiona tree (2022-09-09T12Z, the peak-activity IC within the Fiona precursor environment) seeds near 19°N in the central Atlantic and fans westward into the Caribbean, consistent with Fiona’s observed track. The Ian tree (2022-09-22T00Z, 24 hours before Ian’s NHC genesis designation) seeds near 17°N in the western Caribbean and fans northwestward into the Gulf of Mexico, consistent with Ian’s observed track toward Florida.

3.4 Committor Curves and Spatial Genesis Fields

Figure 9 shows the committor curve $p_B(\lambda_i)$ — the probability of eventually reaching state B given arrival at interface λ_i [Rotskoff and Vanden-Eijnden, 2018] — averaged across all 98 initial conditions for both the FFS (AI model) and IFS (brute-force ensemble). The all-season FFS curve shows a near-linear decline with the steepest drop between the first and second shooting interfaces ($\lambda_1 \rightarrow \lambda_2$), consistent with the seasonal mean of P_1 being the lowest of the four conditional probabilities (Table 1). The FFS and IFS curves agree in overall *shape*: both show the steepest committor drop at the same interface level, and both identify the same rate-limiting stage of the intensification cascade. This structural agreement — across two completely independent modelling systems — provides suggestive cross-model consistency that the intensification bottleneck identified by FFS reflects real atmospheric physics rather than artefacts of the SDL-WXFormer architecture; we note that the comparison is complicated by the different ensemble generation mechanisms (learned stochastic layers vs. singular-vector perturbations) and the 1° vs. 1.5° resolution asymmetry, so quantitative agreement is not expected and not claimed. The FFS committors are systematically higher in *magnitude*, consistent with SDL-WXFormer’s higher absolute genesis rate at 1° relative to IFS archived at 1.5° (the coarser IFS output smooths MSLP minima, effectively raising the intensification barrier).

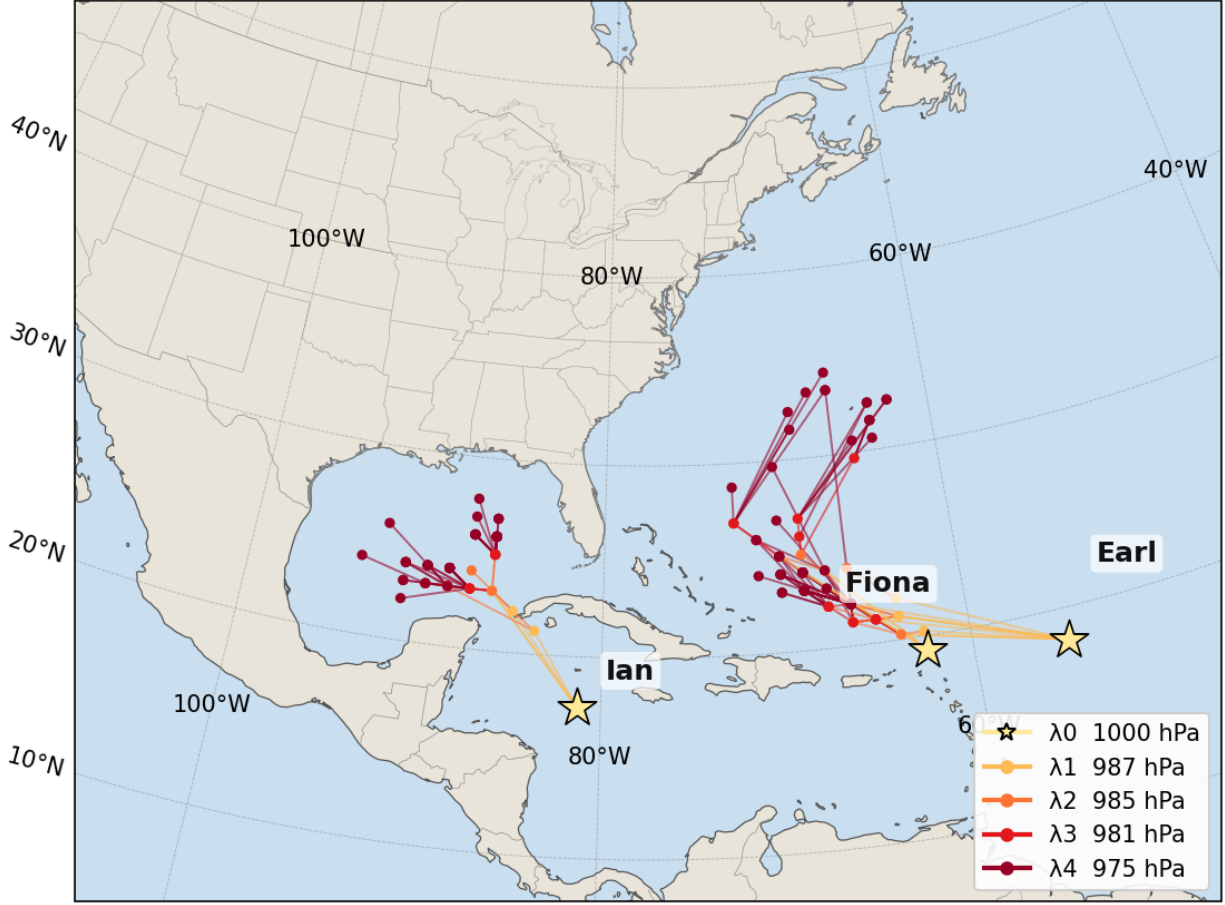


Figure 8: FFS shooting trees for the three case-study ICs on a single combined map. Each star (λ_0 seed) and its descendant branches are plotted together; branch colour encodes interface level from λ_1 (yellow-orange) to λ_4 (dark red), using the same colour scale for all three ICs. IC labels identify each seed: Earl (2022-09-02T00Z, seed near 18°N in the eastern Atlantic, 18 hours before Earl’s genesis), Fiona (2022-09-09T12Z, seed near 19°N in the central Atlantic), and Ian (2022-09-22T00Z, seed near 17°N in the western Caribbean). Earl trajectories fan westward consistent with Earl’s observed west-northwest track; Fiona trajectories track westward into the Caribbean consistent with Fiona’s subsequent path; Ian trajectories fan northwestward into the Gulf of Mexico, anticipating Ian’s observed track toward Florida.

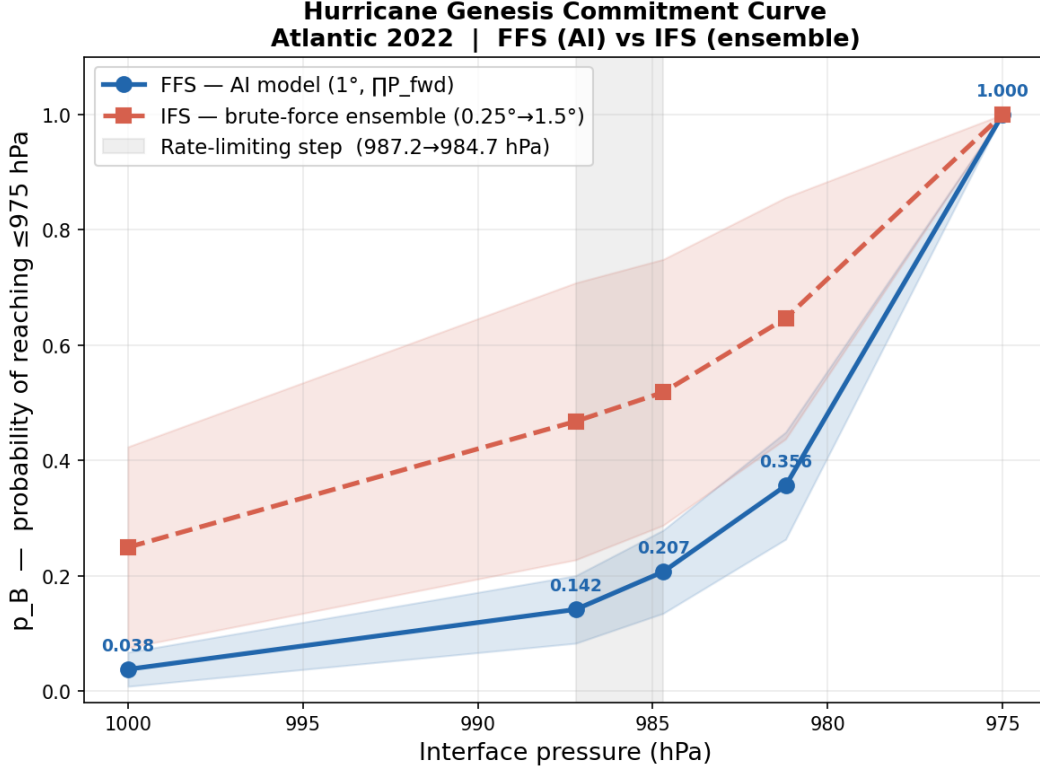


Figure 9: Genesis committor curve $p_B(\lambda_i)$ — the probability of eventually reaching state B (≤ 975 hPa) given arrival at interface λ_i — averaged across all 98 initial conditions. *Blue*: FFS AI model mean $\pm 1\sigma$ across ICs. *Orange*: IFS brute-force ensemble mean $\pm 1\sigma$ across ICs. The shaded bands show IC-to-IC variability; the annotated values are the all-season FFS means. The grey band highlights the rate-limiting step (largest single drop), which falls between the first and second shooting interfaces for the full 2022 Atlantic season. The FFS and IFS curves agree in the overall shape while differing in magnitude, consistent with the resolution and physics differences between the two systems.

3.5 Physics Along a Reactive Pathway

Figure 10 traces the storm-centred thermodynamic and kinematic evolution along a representative Earl reactive pathway (pathway 1079) from λ_0 through state B . The vortex originates near 16°N in the eastern Atlantic and tracks west-northwestward to 28°N as it intensifies from ~ 999 hPa to 973 hPa, consistent with Earl’s observed tropical Atlantic genesis and subsequent westward motion. The persistent cyclonic vorticity (ζ_{850}) and warm-core anomaly (T'_{500}) at each interface confirm continuous vortex identity and warm-core structure through state B . These storm-centred panels reveal the physical mechanisms operating at each stage, connecting the abstract conditional probabilities in Table 1 to concrete dynamical processes.

3.6 Storm-Centred Physics Composites

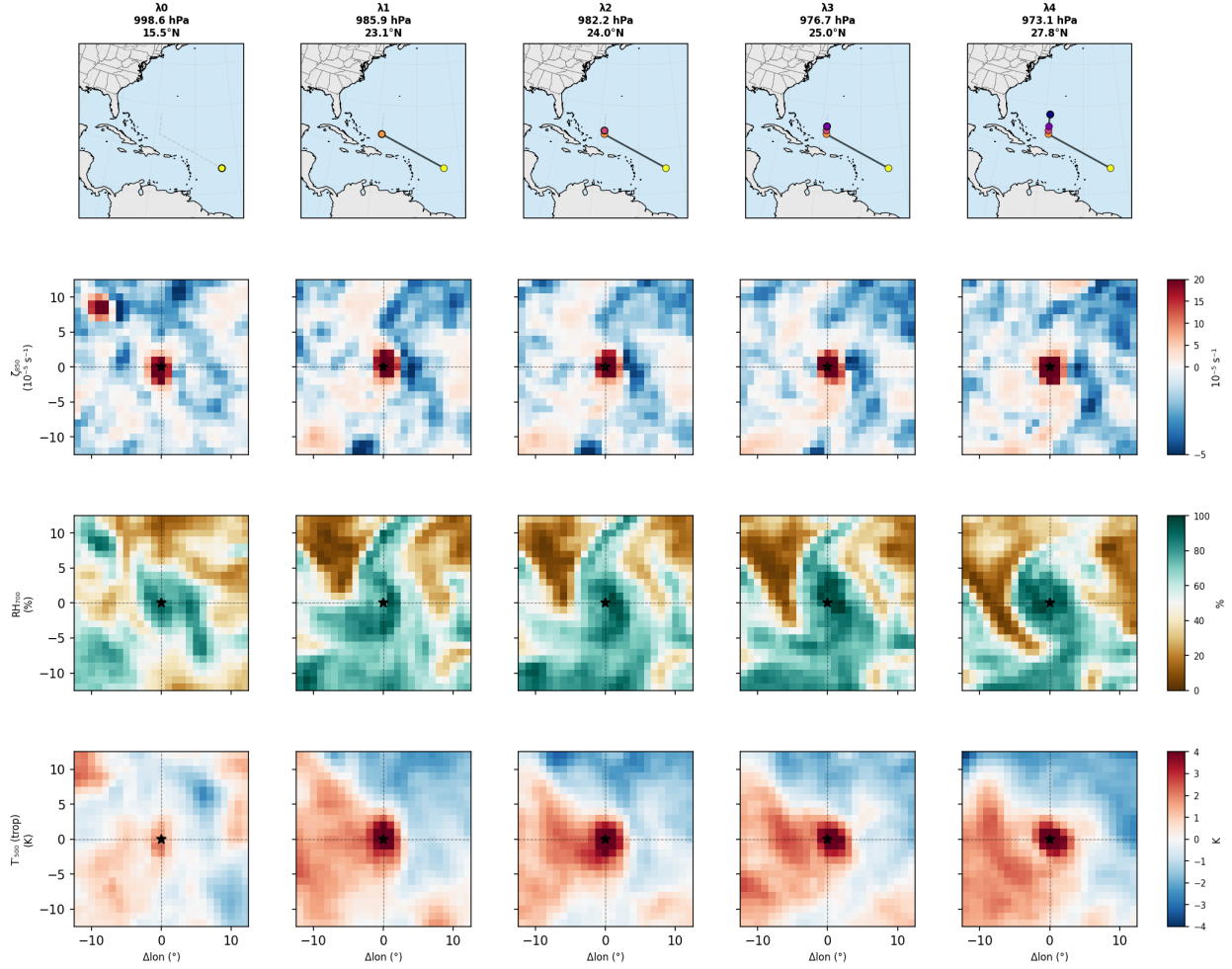
Figures 11–13 show storm-centred composites of vorticity, relative humidity, and warm-core anomaly, composited across all reactive pathways for each case-study IC. By averaging over all configurations that successfully reach state B , these composites reveal the mean thermodynamic and kinematic environment that supports genesis, smoothing out pathway-to-pathway variability to expose the underlying physical signal.

All three ICs share a common structural signature: a compact positive vorticity core at 850 hPa, a saturated inner core ($\text{RH}_{700} \gtrsim 70\%$), and a growing warm-core anomaly at 500 hPa that strengthens progressively from λ_0 through state B . The amplitude of T'_{500} at state B is comparable across the three cases despite their differing genesis rates, suggesting that once a vortex reaches state B it has traversed a physically similar intensification pathway regardless of the environmental background. The primary inter-IC difference appears at early interfaces: the Earl IC composites show weaker vorticity and warm-core organisation at λ_1 – λ_2 compared to the Fiona and Ian composites, directly reflecting the lower P_1 value for Earl (Table 1) and confirming that the conditional probability bottleneck has a physically interpretable composite signature.

4 Discussion and Conclusions

4.1 Methodological Validation and Practical Considerations

The central self-consistency result — FFS and direct-sampling rates agreeing within 1.03 ± 0.15 across 98 initial conditions — confirms the correct implementation of the conditional probability factorization. These estimates use separate components of the same simulation: the FFS rate uses only the interface crossing counts and shooting probabilities, while the direct rate uses only the raw state- B arrival counts from flux trajectories. Their close agreement across three orders of magnitude of genesis probability confirms that (1) the interface spacing produces well-conditioned conditional probability estimates, (2) the ensemble sizes ($N = 2000$ per interface) are sufficient, and (3) the CPS contamination screen does not introduce systematic bias into the factorization. As discussed in Section 3.1, this self-consistency check does not constitute external validation against



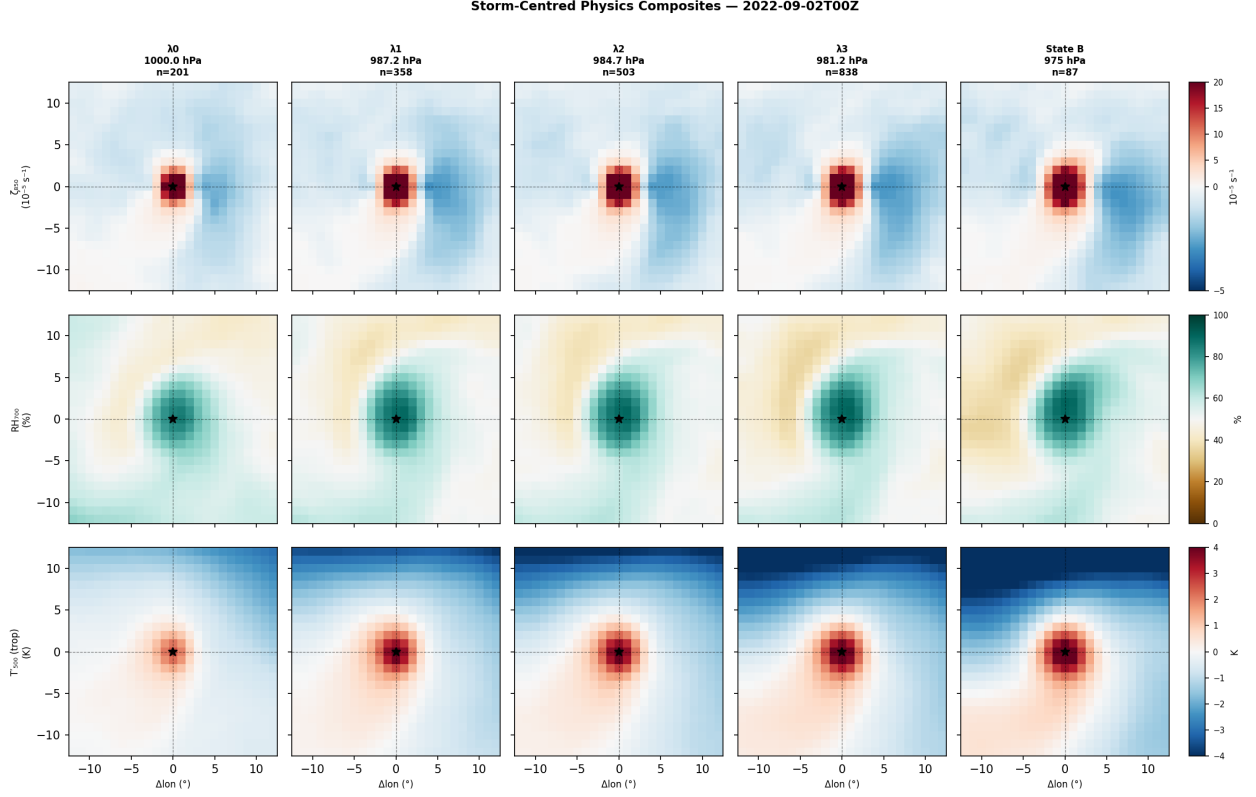


Figure 11: Storm-centred physics composites along reactive pathways for the Earl IC (2022-09-02T00Z). Columns show interfaces λ_0 through state B ; sample sizes n in each column header. Rows show: (1) 850-hPa relative vorticity ζ_{850} (10^{-5} s^{-1}); (2) 700-hPa relative humidity RH_{700} (%); (3) 500-hPa warm-core anomaly T'_{500} (K). The weaker vorticity and warm-core organisation at λ_1 reflects Earl's lower P_1 value and identifies initial organisation as the rate-limiting bottleneck.

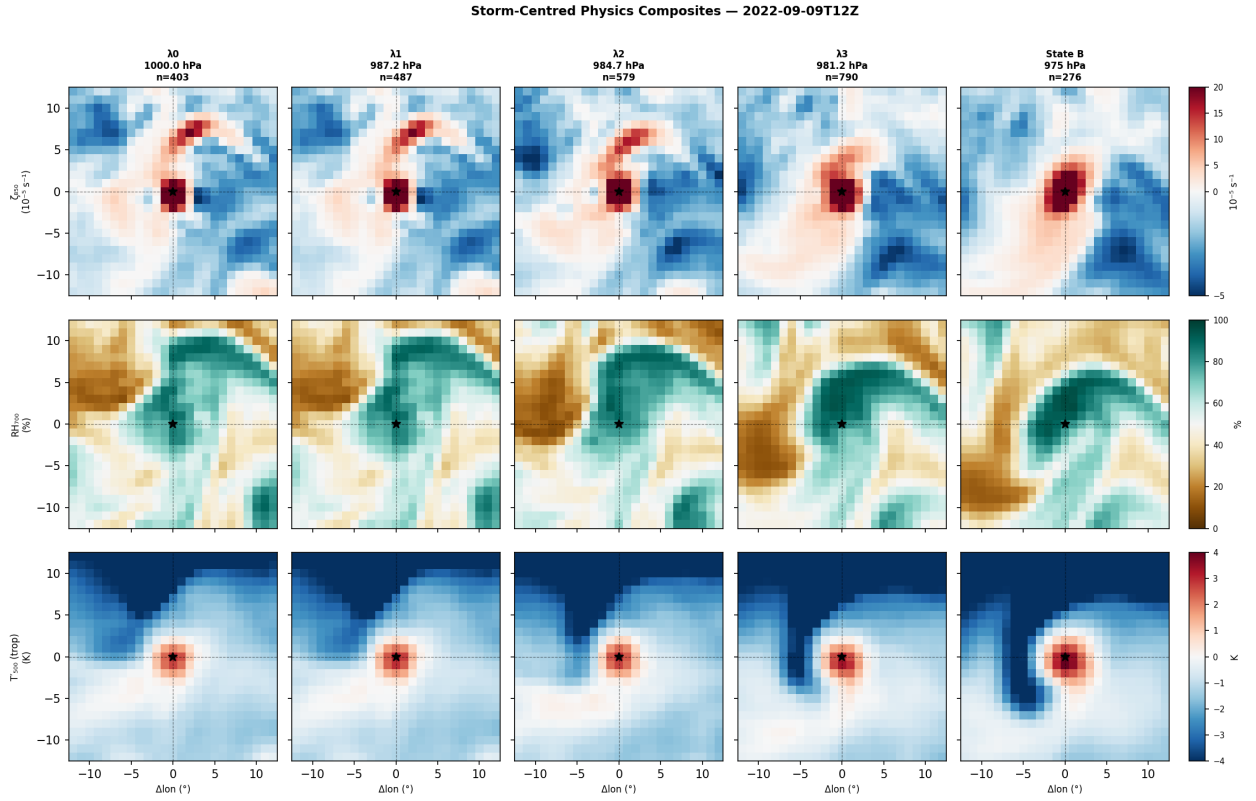


Figure 12: As in Fig. 11 but for the Fiona IC (2022-09-09T12Z). Stronger vorticity and warm-core development at early interfaces compared to Earl reflects the more favourable large-scale environment and higher P_1 .

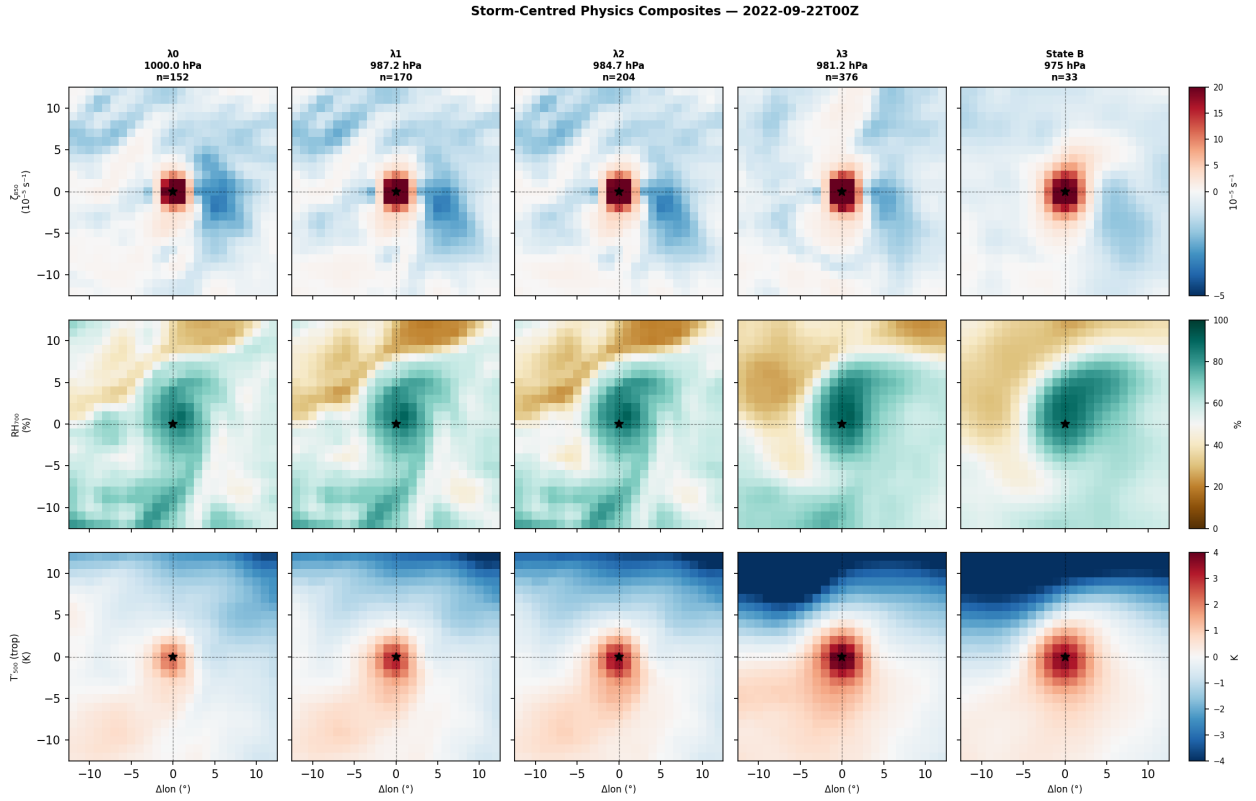


Figure 13: As in Fig. 11 but for the Ian IC (2022-09-22T00Z). The T'_{500} amplitude at state B is comparable across all three ICs despite their differing genesis rates, suggesting a physically similar final intensification pathway once the vortex reaches the near-hurricane threshold.

observed climatology, nor does it test the Markov assumption; it confirms that the FFS bookkeeping arithmetic is implemented correctly within the SDL-WXFormer framework.

Enhancement factors ($3\text{--}140\times$, geometric mean $14\times$) scale inversely with genesis probability, as expected from FFS theory [Allen et al., 2009]. The right-skewed speedup distribution mirrors the log-normal FFS rate distribution itself; the geometric mean is the appropriate summary for such distributions and is reported throughout. For the most active environments where direct sampling is feasible (speedup $3\times$), the two methods serve as mutual validation. For suppressed environments (speedup $\sim 100\times$), FFS is the only practical route to a quantitative genesis rate estimate.

The bottleneck analysis adds physical content beyond a scalar rate. For the Earl IC, the rate-limiting step at λ_1 (initial organization) indicates that the environment is resistant to producing organized convection but that once organized, the system is likely to reach hurricane strength. For the Fiona IC, the uniformly elevated probabilities with a $P_4 = 0.276$ terminal barrier indicate an environment that readily supports each successive intensification step but where crossing the final 975 hPa threshold requires a specific convective alignment not guaranteed even in this highly active environment. For the Ian IC, the easy initial organization but hard final intensification ($P_4 = 0.102$) suggests an environment whose large-scale structure is favorable for genesis attempts but where the thermodynamic pathway to full hurricane development requires specific mesoscale alignment not yet present 24 hours before NHC designation. These MSLP-based bottleneck identifications are corroborated by independent thermodynamic and kinematic evidence: the storm-centred composites (Figs 11–13) show that the interface at which each IC’s P_i is smallest corresponds to a measurably weaker vorticity core, reduced mid-level relative humidity, and less developed warm-core anomaly in the ensemble-mean reactive pathway, confirming that the pressure-based bottleneck reflects a genuine physical barrier in the intensification cascade.

The CPS filter warrants clarification. At 1° resolution, formal CPS parameters are not quantitatively validated against higher-resolution analyses, and our implementation omits the thermal asymmetry (B) parameter. The filter is best understood as a contamination screen rather than a rigorous warm-core classifier — rejecting obviously extratropical systems while accepting all ambiguous cases. Future implementations with finer-resolution models could apply a more principled multidimensional CPS criterion.

4.2 Limitations

This study analyzes a single hurricane season (2022). Interannual variability driven by ENSO, the AMO, and Saharan dust outbreaks is not represented. The SDL-WXFormer was trained on ERA5 (1979–2021) at 1° resolution, which smooths intense vortex cores and underrepresents Category 3+ hurricanes; state B is accordingly defined at 975 hPa rather than the intensities typical of major hurricane activity. Validation against physics-based models (WRF, IFS) at higher resolution is an important next step.

Additionally, MSLP as an order parameter has known limitations at 1° resolution: terrain-induced pressure artifacts (present in this model due to topographic interpolation differences) re-

quire the CPS filter and geographic masking described in Section 2.3. MSLP is a downstream integrator of the thermodynamic and kinematic preconditions for genesis — warm-core development, vorticity alignment, mid-level moistening, shear suppression — rather than a direct measure of those precursors. As a result, the FFS rate strictly measures the frequency of organised MSLP deepening, and the per-interface crossing probabilities reflect barriers to pressure deepening specifically. That said, the storm-centred composite figures (Figs 11–13) show that the MSLP-defined interfaces correspond to coherent and physically interpretable thermodynamic states: vorticity, mid-level relative humidity, and warm-core anomaly all evolve progressively and consistently from λ_0 through state B , validating that the MSLP interfaces are tracking genuine tropical development rather than spurious pressure minima. Nonetheless, reaction coordinates that incorporate vorticity, saturation fraction, or a composite genesis potential index [e.g., Bister and Emanuel, 1998, Camargo et al., 2007] are expected to sharpen the physical interpretability of per-interface bottleneck diagnostics and are a natural extension for future work.

4.3 Conclusions

Forward Flux Sampling resolves Atlantic tropical cyclogenesis rates spanning three orders of magnitude from 98 atmospheric initial conditions in a fast neural weather emulator, with a self-consistency check confirming FFS and direct-sampling rates agree within 3%. Computational enhancement of $3\text{--}140\times$ (geometric mean $14\times$) makes FFS practical across the full range of synoptic environments. Physically, the method reveals that the rate-limiting step in genesis varies by regime: initial tropical organization (P_1 bottleneck, as in Earl), uniformly elevated crossing probabilities with a terminal intensification barrier (P_4 bottleneck, as in the Fiona IC), or a compound barrier at the final intensification stages ($P_3\text{--}P_4$, as in the Ian IC). These distinct bottleneck signatures encode mechanistic information about genesis pathway dynamics that is invisible to scalar rate estimates from direct sampling. A cross-model comparison of genesis committor curves between SDL-WXFormer and ECMWF IFS shows suggestive consistency in shape and rate-limiting stage (Fig. 9), providing qualitative evidence that the bottleneck structure is not an artefact of the neural emulator’s dynamics.

The methodology extends directly to other atmospheric rare events (extreme heatwaves, sudden stratospheric warmings, blocking onset) and to physics-based models. Any generative weather or climate model that produces an ensemble of plausible atmospheric trajectories from a shared initial state — whether a diffusion-based emulator, a flow-matching model, or a stochastic parameterization scheme — satisfies the architectural Markov requirement and can in principle be coupled to FFS in the same way demonstrated here. The combination of neural emulators with rare event sampling therefore opens a broadly applicable and computationally tractable path to systematic characterization of low-probability, high-consequence atmospheric events across climatic regimes.

A Supplementary Information: State- B Threshold Sensitivity

The choice of $\lambda_B = 975$ hPa was determined by the geographic and physical character of state- B arrivals, not solely by computational considerations. Complete simulations were first carried out with $\lambda_B = 965$ hPa. At that threshold, state- B arrivals were disproportionately located at latitudes $> 40^\circ\text{N}$, with many reaching state B via extratropical deepening — including cases analogous to Fiona’s eventual high-latitude track — rather than through tropical intensification. Although the CPS warm-core filter rejects clearly extratropical trajectories, at 965 hPa the filter struggled to cleanly separate borderline recurving storms from genuine tropical development, leading to state- B arrival distributions inconsistent with observed Atlantic genesis climatology.

Raising λ_B to 975 hPa substantially shifted state- B arrivals into the tropical and subtropical Atlantic, Caribbean, and Gulf of Mexico (Fig. 6), consistent with where September Atlantic storms actually develop and track. At this threshold the CPS filter operates more reliably, and the geographic distribution of arrivals matches physical expectations for the 2022 season. A secondary benefit was that P_4 values, which had collapsed below 0.05 across most ICs at 965 hPa (concentrating nearly all variance in the final interface), distributed more uniformly at 975 hPa, maintaining the well-conditioned range targeted by the interface placement algorithm.

The shooting interface values (1000.0, 987.2, 984.7, 981.2 hPa) were optimised using the 965 hPa simulations as a starting point: the crossing statistics from those longer runs informed the optimal interface spacing for the 975 hPa state B , and the final values were rounded to physically interpretable pressure levels.

The sensitivity of reported genesis *rates* to threshold choice is expected: $k(\lambda_B = 975)$ and $k(\lambda_B = 965)$ differ by roughly one order of magnitude, consistent with the additional conditional probability factor for the extra interface. The key methodological properties — self-consistency, seasonal cycle structure, and bottleneck diagnostics — are qualitatively robust across thresholds. For physics-based models at finer resolution, a lower λ_B (e.g., 960 hPa) may be more appropriate, and the interface placement should be re-optimised accordingly.

Acknowledgments

The author thanks colleagues at NCAR’s Earth System Data Science group for discussions on neural weather prediction. Computational resources provided by the NCAR-Wyoming Supercomputing Center (Derecho and Casper systems) supported this research. This work was supported by NSF grants AGS-XXXXXX and CSSI-XXXXXX.

Data Availability Statement

The FFS driver code, vortex tracker, and contamination screen are available at <https://github.com/NCAR/miles-tails>. Note that `config/ffs.yml` in the repository reflects a development-phase parameter set; the operative interface values and thresholds are those stated in Section 2.1 of this

paper. The FFS simulation output (rate statistics, conditional crossing probabilities, and reactive pathway summaries for all 98 initial conditions) will be archived at the NCAR Research Data Archive upon acceptance. ERA5 reanalysis initial conditions [Hersbach et al., 2020] are available from the Copernicus Climate Data Store (<https://cds.climate.copernicus.eu>). IFS ensemble data are available through WeatherBench2 [Rasp and Coauthors, 2024]. **[SDL-WXFormer model weights: update repository link before submission.]**

References

- R. J. Allen, D. Frenkel, and P. R. ten Wolde. Simulating rare events in equilibrium or nonequilibrium stochastic systems. *J. Chem. Phys.*, 124:024102, 2006.
- R. J. Allen, C. Valeriani, and P. R. ten Wolde. Forward flux sampling for rare event simulations. *J. Phys.: Condens. Matter*, 21:463102, 2009. doi: 10.1088/0953-8984/21/46/463102.
- Rosalind J. Allen, Patrick B. Warren, and Pieter Rein ten Wolde. Sampling rare switching events in biochemical networks. *Physical Review Letters*, 94:018104, 2005. doi: 10.1103/PhysRevLett.94.018104.
- K. Bi and Coauthors. Accurate medium-range global weather forecasting with 3D neural networks. *Nature*, 619:533–538, 2023. doi: 10.1038/s41586-023-06185-3.
- M. Bister and K. A. Emanuel. Dissipative heating and hurricane intensity. *Meteor. Atmos. Phys.*, 65:233–240, 1998. doi: 10.1007/BF01030791.
- P. G. Bolhuis, D. Chandler, C. Dellago, and P. L. Geissler. Transition path sampling: Throwing ropes over rough mountain passes, in the dark. *Annu. Rev. Phys. Chem.*, 53:291–318, 2002.
- R. Buizza and Coauthors. The evolution of ensemble forecasting systems. *Bull. Amer. Meteor. Soc.*, 99:1393–1400, 2018.
- S. J. Camargo, K. A. Emanuel, and A. H. Sobel. Use of a genesis potential index to diagnose ENSO effects on tropical cyclone genesis. *J. Climate*, 20:4819–4834, 2007. doi: 10.1175/JCLI4282.1.
- J. Finkel, E. P. Gerber, D. S. Abbot, and J. Weare. Revealing the statistics of extreme events hidden in short weather forecast data. *AGU Adv.*, 4:e2023AV000881, 2023. doi: 10.1029/2023AV000881.
- J. Finkel, D. S. Abbot, and J. Weare. Bringing it all together: Rare event sampling in climate science. *Chaos*, 34:072101, 2024.
- N. L. Frank. Atlantic tropical systems of 1969. *Mon. Wea. Rev.*, 98:307–314, 1970.
- V. M. Gálfi, V. Lucarini, and J. Wouters. A large deviation theory-based analysis of heat waves and cold spells in a simplified model of the general circulation of the atmosphere. *J. Stat. Mech.: Theory Exp.*, 2019:033404, 2019. doi: 10.1088/1742-5468/ab02e8.

- W. M. Gray. Global view of the origin of tropical disturbances and storms. *Mon. Wea. Rev.*, 96: 669–700, 1968.
- D. J. Halperin, H. E. Fuelberg, R. E. Hart, J. H. Cossuth, P. Sura, and R. J. Pasch. An evaluation of tropical cyclone genesis forecasts from global numerical models. *Wea. Forecasting*, 28:1423–1445, 2013. doi: 10.1175/WAF-D-13-00008.1.
- R. E. Hart. A cyclone phase space derived from thermal wind and thermal asymmetry. *Mon. Wea. Rev.*, 131:585–616, 2003.
- M. Heikenfeld and Coauthors. tobac 1.2: towards a flexible framework for tracking and analysis of clouds in diverse datasets. *Geosci. Model Dev.*, 12:4551–4570, 2019.
- H. Hersbach, B. Bell, P. Berrisford, S. Hirahara, A. Horányi, J. Muñoz-Sabater, J. Nicolas, C. Peubey, R. Radu, D. Schepers, A. Simmons, C. Soci, S. Abdalla, X. Abellan, G. Balsamo, P. Bechtold, G. Biavati, J. Bidlot, M. Bonavita, G. De Chiara, P. Dahlgren, D. Dee, M. Diamantakis, R. Dragani, J. Flemming, R. Forbes, M. Fuentes, A. Geer, L. Haimberger, S. Healy, R. J. Hogan, E. Hólm, M. Janisková, S. Keeley, P. Laloyaux, P. Lopez, C. Lupu, G. Radnoti, P. de Rosnay, I. Rozum, F. Vamborg, S. Villaume, and J.-N. Thépaut. The ERA5 global reanalysis. *Quart. J. Roy. Meteor. Soc.*, 146:1999–2049, 2020. doi: 10.1002/qj.3803.
- S. Hussain and A. Haji-Akbari. Studying rare events using forward-flux sampling: Recent breakthroughs and future outlook. *J. Chem. Phys.*, 152:060901, 2020. doi: 10.1063/1.5127780.
- J. A. Knaff and R. M. Zehr. Reexamination of tropical cyclone pressure–wind relationships. *Wea. Forecasting*, 22:71–88, 2007.
- A. Laio and M. Parrinello. Escaping free-energy minima. *Proc. Natl. Acad. Sci.*, 99:12562–12566, 2002.
- R. Lam and Coauthors. Learning skillful medium-range global weather forecasting. *Science*, 382: 1416–1421, 2023. doi: 10.1126/science.adi2336.
- A. Lancelin, A. Wikner, L. Dubus, C. Le Priol, D. S. Abbot, F. Bouchet, P. Hassanzadeh, and J. Weare. AI-boosted rare event sampling to characterize extreme weather. *arXiv preprint arXiv:2510.27066*, 2025.
- C. W. Landsea. Atlantic basin hurricanes: Indices of climatic changes. *Climatic Change*, 42:89–129, 1998.
- E. N. Lorenz. The predictability of a flow which possesses many scales of motion. *Tellus*, 21: 289–307, 1969.
- B. Mu, Y. Chen, S. Yuan, B. Qin, and H. Guo. Skillful subseasonal-to-seasonal forecasting of extreme events with a multi-sphere coupled probabilistic model. *arXiv preprint arXiv:2512.12545*, 2025.

- National Hurricane Center. 2022 Atlantic hurricane season. Technical report, NOAA/National Hurricane Center, 2023. URL https://www.nhc.noaa.gov/data/tcr/summary_2022.pdf. Annual Tropical Cyclone Report.
- J. Pathak and Coauthors. FourCastNet: A global data-driven high-resolution weather model using adaptive Fourier neural operators. *arXiv:2202.11214*, 2022.
- D. A. Plotkin, R. J. Webber, M. E. O’Neill, J. Weare, and D. S. Abbot. Maximizing simulated tropical cyclone intensity with action minimization. *J. Adv. Model. Earth Syst.*, 11:863–891, 2019.
- F. Ragone, J. Wouters, and F. Bouchet. Computation of extreme heat waves in climate models using a large deviation algorithm. *Proc. Natl. Acad. Sci.*, 115:24–29, 2018. doi: 10.1073/pnas.1712645115.
- S. Rasp and Coauthors. WeatherBench 2: A benchmark for the next generation of data-driven global weather models. *J. Adv. Model. Earth Syst.*, 16:e2023MS004019, 2024. doi: 10.1029/2023MS004019.
- G. M. Rotskoff and E. Vanden-Eijnden. Neural networks as optimal approximations of committor functions for Markov processes. *NeurIPS Workshop on Integration of Deep Learning Theories*, 2018.
- J. R. Schreck and Coauthors. SDL-WXFormer: Stochastic decomposition layers for controllable weather ensemble forecasting. *arXiv:2512.18815*, 2025.
- John S. Schreck, Thomas E. Ouldrige, Flavio Romano, Petr Šulc, Liam P. Shaw, Ard A. Louis, and Jonathan P. K. Doye. DNA hairpins destabilize duplexes primarily by promoting melting rather than by inhibiting hybridization. *Nucleic Acids Res.*, 43:6181–6190, 2015. doi: 10.1093/nar/gkv582.
- D. W. Titley and R. L. Elsberry. Large intensity changes in tropical cyclones: A climatology and case studies for the western North Pacific. *Mon. Wea. Rev.*, 128:92–111, 2000.
- H. Touchette. The large deviation approach to statistical mechanics. *Phys. Rep.*, 478:1–69, 2009.
- C. Valeriani, R. J. Allen, M. J. Morelli, D. Frenkel, and P. R. ten Wolde. Computing stationary distributions in equilibrium and nonequilibrium systems with forward flux sampling. *J. Chem. Phys.*, 127:114109, 2007. doi: 10.1063/1.2767625.
- R. J. Webber, D. A. Plotkin, M. E. O’Neill, D. S. Abbot, and J. Weare. Practical rare event sampling for extreme mesoscale weather. *Chaos*, 29:053109, 2019.